

REVIEW

After the last Ozempic shot, the old biology starts voting again

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What happens when semaglutide stops? For most people, the drug's appetite and glucose effects fade, weight begins to return, and some metabolic gains shrink — but the speed and extent are far from fixed.

In one of the clearest withdrawal experiments, 803 adults first spent 20 weeks taking semaglutide. They had lost about a tenth of their starting weight when the investigators split them: two-thirds kept injecting; one-third received placebo. Over the next 48 weeks, the first group kept losing while the second gained 6.9%.¹

That split makes the question sharper than “does dieting work?” Both groups still received lifestyle support. What changed was exposure to a [GLP-1 receptor agonist](#), a drug that mimics a gut signal involved in appetite and blood-sugar regulation.^{2,3,4} The evidence says stopping usually removes an active brake. It does not reveal an established addiction-like withdrawal syndrome.^{5,6,7}

01 THE SIGNAL FADES BEFORE THE STORY IS FINISHED

Semaglutide stays in the body for weeks, so the transition is usually a slope rather than a cliff. As drug exposure falls, its effects on fullness, food intake, stomach function, and glucose control fade too.^{2,3} Researchers reviewing discontinuation describe the returning pressure in familiar terms: hunger rises, the body defends its earlier energy stores, and the behaviours that felt easier on treatment may demand effort again.^{5,6}

What would you notice first? The studies are surprisingly bad at answering that in sequence. They measured weight much more reliably than hunger, cravings, meal size, or the day nausea stopped. The honest whole-body picture in [Figure 1](#) therefore separates what trials measured from what pharmacology makes plausible.^{8,9,7}

The name on the pen matters here. The cleanest stopping studies used semaglutide 2.4 mg in adults treated for obesity and excluded diabetes.^{1,8} Trials of injectable semaglutide in type 2 diabetes often used 0.5 or 1.0 mg in people with very different cardiovascular risk.¹⁰ “Stopping Ozempic” can therefore describe several experiments at once: ending glucose-lowering treatment, ending obesity treatment, or ending an off-label course at another dose. The broad direction may travel across those settings. The exact size of the rebound may not.

The final injection is also not the instant when exposure becomes zero. Semaglutide's long residence in the body spreads the handover across several weeks, while trial visits usually sample weight and laboratory values much farther apart.^{2,3} That mismatch helps explain why the evidence can describe a broad rebound yet cannot name the first day hunger, glucose, or digestive symptoms will change for one person.

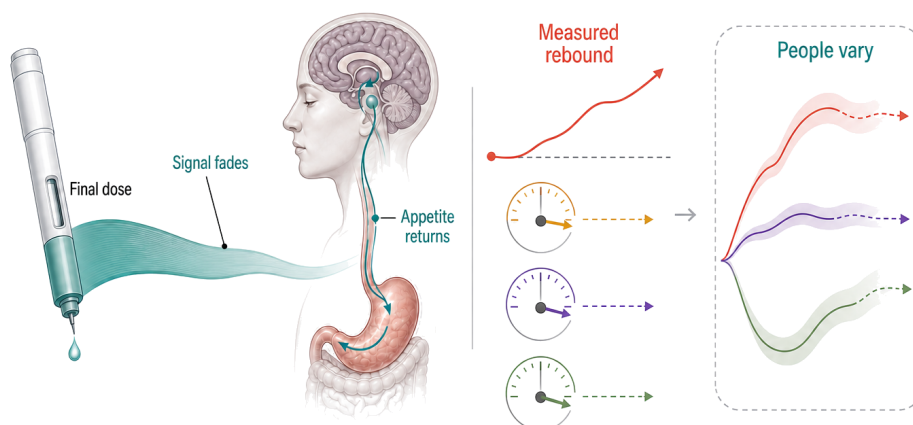


Figure 1. The drug's signal fades; the body's older pressures remain. Follow the path from the final dose to weaker appetite and glucose effects, then to the outcomes researchers actually tracked. The last boundary stays open because group averages cannot forecast one body, and post-stop appetite has been measured less directly than weight.^{1,8,9,5}

02 WEIGHT RETURNS IN MONTHS, NOT ALL AT ONCE

Wilding and colleagues followed 327 participants for a year after STEP 1 ended. Those who had taken semaglutide 2.4 mg had lost 17.3% of their starting weight; a year after treatment and structured lifestyle support stopped, they had regained 11.6 percentage points — roughly two-thirds of the loss.⁸ They still averaged 5.6% below baseline. “Regain” did not mean instant erasure.

Those numbers answer two questions that are easy to blur. The 11.6-point rise describes how far the group moved upward after treatment; the remaining 5.6% below baseline describes where the group ended relative to where it began. Neither says that every participant regained two-thirds. An average can combine people who regained nearly everything, people who held much of the loss, and people who left the study before the last measurement.

The comparison with STEP 4 adds an important control. In STEP 1, both semaglutide and the trial's structured lifestyle program ended together, so their contributions to the rebound cannot be cleanly separated.⁸ In STEP 4, both randomized groups continued lifestyle support while only medication exposure changed: the placebo-switched group gained 6.9% as the continuing group lost another 7.9%.¹ The two studies therefore point in the same direction through different designs, while leaving longer-term maintenance unanswered.

When another team pooled eight randomized trials, people stopping semaglutide or tirzepatide regained an estimated 9.69 kg, though the interval was wide enough to signal major differences among studies.¹¹ A newer reconstruction estimated about one kilogram per month during observed follow-up, but its return-to-baseline projection extended beyond the available 52-week data.¹² The evidence-bearing estimates appear in [Figure 2](#); the modeled value is identified by its source label, and none is an individual forecast.

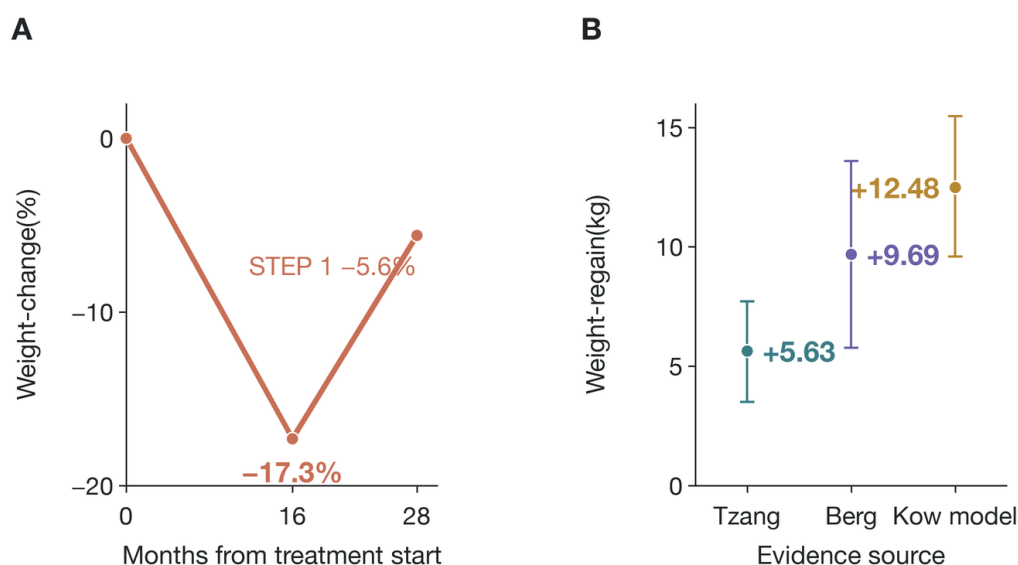


Figure 2. Most lost weight returned, but the estimates do not form one universal curve. Panel A follows percentage weight change from STEP 1 baseline through treatment and one year off. Panel B names three evidence sources as categories and places their kilogram regain estimates, with attached intervals, on one vertical scale. “Kow model” is arithmetic scaling of a monthly estimate, not an observed 12-month mean.^{8,11,12,9}

03 THE REST OF THE METABOLIC DASHBOARD MOVES TOO

Weight was not the only line that bent back. In the STEP 1 extension, waist size, blood pressure, blood fats, and measures of glucose regulation generally drifted toward their pretreatment values as weight returned.⁸ Across 18 randomized trials, a team led by Tzang estimated that stopping GLP-1 treatment in people with obesity raised HbA1c, the roughly three-month blood-sugar average, by

0.25 percentage points; waist circumference, fasting glucose, and systolic blood pressure also worsened.⁹

The diabetes story carries higher stakes and thinner semaglutide-specific evidence. In pooled studies of people with type 2 diabetes, HbA1c rose by 0.65 points after GLP-1 therapy stopped, but the studies mixed drugs and produced very inconsistent estimates.⁹ Trials show that semaglutide can improve cardiovascular outcomes during treatment.^{10,13,14} They do not tell us that cardiovascular protection vanishes on the date of the final injection.

EVIDENCE WINDOW	POPULATION AND DESIGN	WHAT CHANGED AFTER STOPPING	WHAT IT CANNOT TELL YOU	SOURCE
STEP 4 withdrawal	803 adults, no diabetes	+6.9% off drug; -7.9% continuing	Years of use; diabetes doses	¹
STEP 1 extension	327 adults	11.6 points regained; markers drifted back	Drug stop apart from support stop	⁸
18-trial meta-analysis	Obesity and diabetes; mixed drugs	Obesity +5.63 kg; diabetes HbA1c +0.65 points	Personal timing; heterogeneity was high	⁹
Supported follow-up	25 women with <u>polycystic ovary syndrome</u>	About one-third regained	Whether metformin or support helped	¹⁵

04 THE AVERAGE CURVE HIDES THE MOST IMPORTANT EXCEPTION

The pooled number is useful, then dangerous. Tzang's obesity estimate carried heterogeneity above 99%: study

results differed far more than sampling error alone would explain.⁹ A second meta-analysis also found large between-study differences.¹⁶ Dose, drug, time on treatment, initial loss, diabetes, follow-up length, and maintenance support all changed from study to study.^{11,12}

Jensterle's small cohort supplies the complication a tidy rebound story needs. Twenty-five women with obesity and polycystic ovary syndrome stopped semaglutide after 16 weeks but continued metformin and lifestyle support. Two years later they had regained about one-third of the treatment-associated loss, and 21 remained lighter than at baseline.¹⁵ The study was uncontrolled and highly selec-

ted. Still, it kills the certainty: rebound is common, not mechanically complete.

The branching paths in Figure 3 keep that uncertainty visible. They are evidence boundaries, not a menu of proven fixes.^{12,15,17,18}

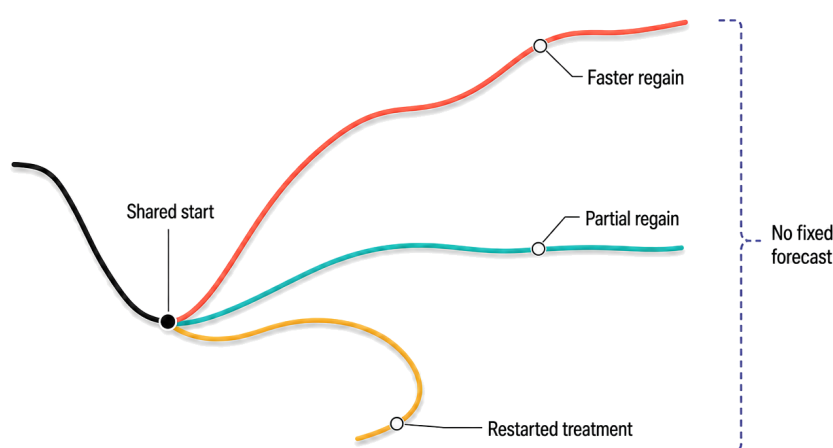


Figure 3. One average rebound contains several different journeys. The shared trunk is the well-supported return of weight after exposure ends. Branches show observed patterns, not assigned futures: a small supported cohort had partial regain without establishing why, and real-world regain often preceded restarting. No branch is a proven strategy or reliable personal prediction.^{15,19,12,20,17}

05 WHY PEOPLE STOP CHANGES WHAT HAPPENS NEXT

Outside trials, a final dose may mark nausea, cost, shortages, a coverage decision, disappointing loss, or a planned pause. Thomsen's review found that 20–50% of real-world users stopped within a year, while adherent users came closer to trial-level results.²¹ In France's early-access program, investigators found a distinct early-discontinuation pattern in 13% of 6,990 semaglutide starters.²² Surveys and cohorts also connect persistence with affordability, support, response, and gastrointestinal symptoms.^{23,24,25}

Those stomach and bowel symptoms are common on treatment. In a 2,549-person observational comparison, about half of semaglutide users reported at least one adverse event, chiefly gastrointestinal.²⁴ Reviews and comparative studies likewise identify nausea and digestive problems as prominent reasons for dose changes or stopping.^{21,26} Yet the literature rarely follows each symptom

after the final dose. It supports “the exposure-related effects usually ease” much better than a precise calendar.

06 STOPPING OFTEN BECOMES RESTARTING

Rodriguez and colleagues tracked 125,474 adults who began a GLP-1 drug. Within a year, 64.8% without diabetes and 46.5% with diabetes had discontinued; among people who stopped, each 1% of regained weight was associated with a roughly 2–3% higher chance of restarting at any point in time.¹⁹ A year after discontinuation, more than a third without diabetes and nearly half with diabetes had restarted.¹⁹

That loop exposes a real-world divide. Some people stop because the body has changed; others because access has. Characterisation studies find that discontinuers are not one clinical type, and pharmacy records cannot fully reveal intention.^{20,21} The result is less a clean endpoint than a series of interrupted treatment episodes.

The loop also cautions against reading a prescription gap as a biological success or failure. A person may still be losing when coverage ends, may pause during illness, or may restart under a different name or dose. Electronic records capture dispensing better than appetite, intention, or affordability.^{19,21} That measurement gap is why real-world studies can tell us how often treatment stops while remaining less certain about what each stop means.

07 WHAT DOES NOT SNAP BACK ON SCHEDULE

Regained weight is not a receipt proving that every tissue has returned to its starting state. During treatment, reduced intake, digestive symptoms, and rapid loss may leave some people vulnerable to low nutrient intake or loss of lean tissue.^{27,28,29} The evidence after stopping is too sparse to say whether muscle or micronutrient status recovers at the same pace as appetite.

Researchers have proposed nutrition, exercise, behavioural care, metformin in selected populations, or lower-dose maintenance as ways to hold more of the loss.^{17,18,30}

None has yet won a strong post-semaglutide head-to-head trial. Even pooled analyses that hint at slower regain with behavioural support have intervals compatible with little effect.¹²

That uncertainty is easy to fill with confident advice, especially because the practical need is immediate. But a plausible plan and a proven plan are different things. The current studies do not establish that tapering prevents rebound, that one diet outperforms another after withdrawal, or that supplements replace careful nutritional assessment.^{12,27,17}

So the final injection is not the ending that the prescription box suggests. The best evidence predicts a gradual return of appetite pressure, weight, and some metabolic risk markers; it does not predict the exact path of one person. The experiment that would change that answer is now obvious: compare abrupt stopping, tapering, continued low-dose treatment, and specified maintenance programs while measuring hunger, glucose, body composition, and clinical outcomes for several years. Until then, the average curve is a compass, not a clock.

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